IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

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Executive Summary

Methodologies

- SpaceX 9 Falcon 9 launch data was collected from a SpaceX API and Wikipedia with the requests library and loaded into data frames with Pandas.
- Pandas and NumPy were used to impute missing values and create a target column for a launch outcome called "Class".
- SQL and Python were used to explore launch variables and find relationships.
- Geographic maps and a dashboard for launch sites were implemented with Folium and Dash Plotly.
- Machine learning with scikit-learn libraries was used to predict a successful first stage Falcon 9 launch.

Results

- A decision tree classifier performed after hyperparameter tuning with GridSearchCV.

Introduction

- **Project background and context**
- SpaceX relies on the reusability of the first stage of Falcon 9 rocket launches, which cost less than launches from competitors.
- This project aims to find the cost of a launch by predicting a successful first stage.
- **Problems you want to find answers**
- What type of location is needed for successful launch conditions?
- Where are the most successful first stage launches? Who produces them?
- How do rocket properties (boosters, payload mass) impact launch conditions?
- What orbits do successful launches take?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  - Collect launch data from a SpaceX API and launch records from Wikipedia
- Perform data wrangling
  - Missing values were imputed, irrelevant columns were dropped
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - A range of classification techniques were hyperparameter tuned with GridSearchCV using cross-validation.

Data Collection

Data Collection Methods

- SpaceX API
- Web scraping

Data to Collect

- Launch stage data
- Launch record data

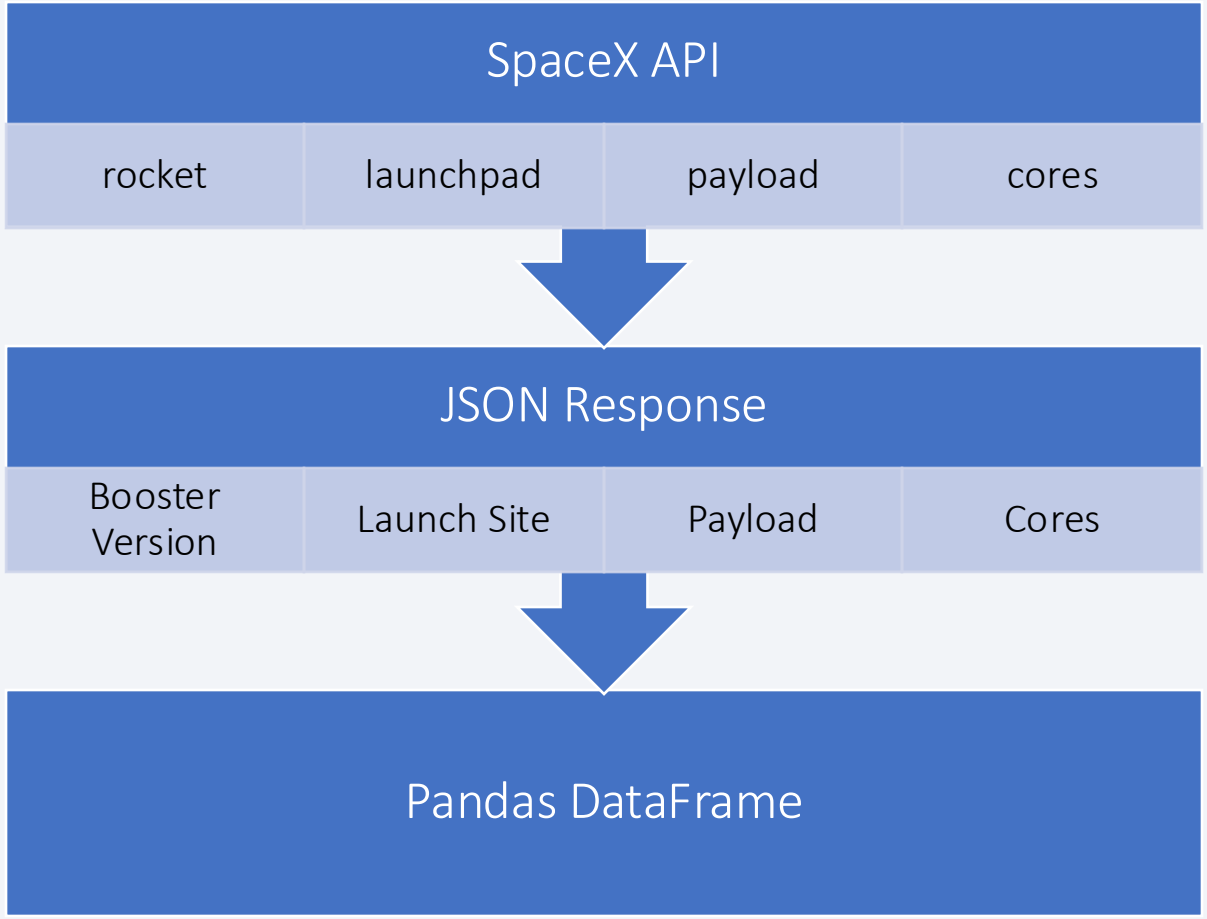
Data Collection – SpaceX API

API Call Data

- rocket: get booster version
- launchpad: get launch site
- payload: get payload mass and orbit
- cores: get outcome of launch

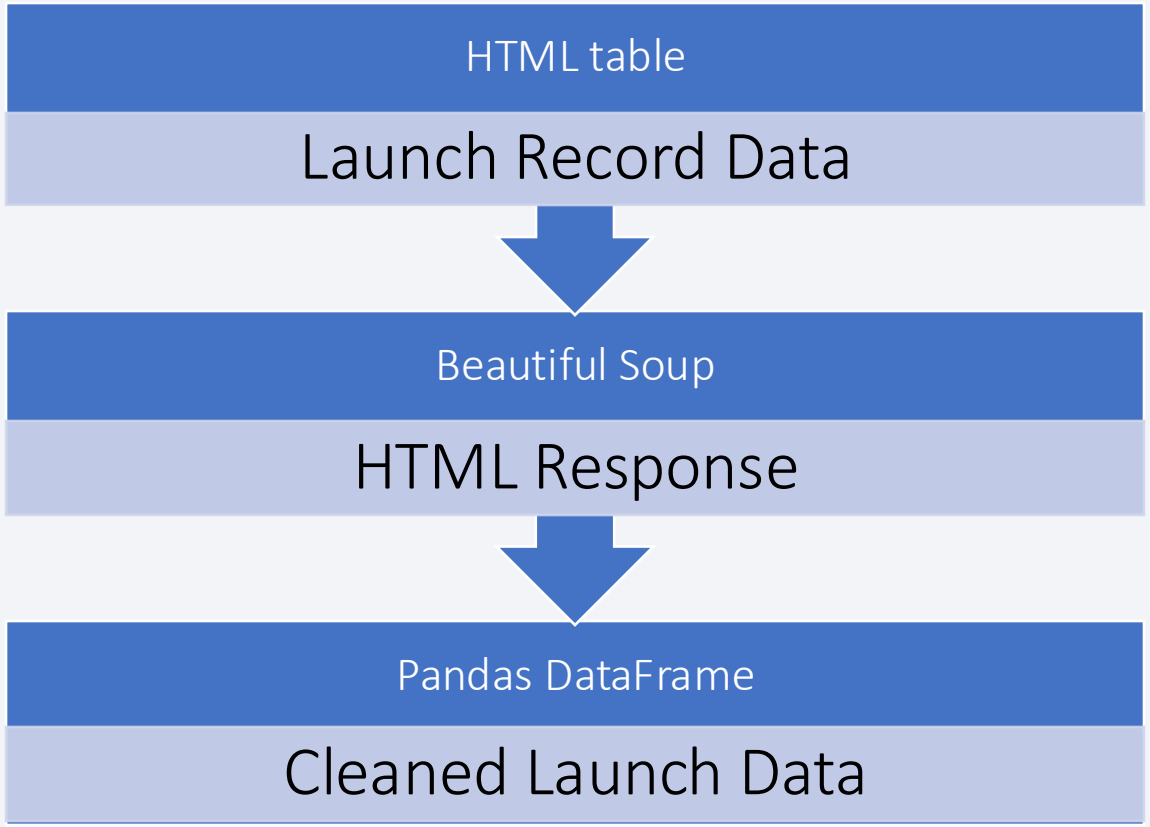
GitHub URL:

https://github.com/Cameron-Nann-Python/ibm_data_science_capstone_rocket_launch_prediction/blob/main/step_1_data_collection/jupyter-labs-spacex-data-collection-api.ipynb



Data Collection - Scraping

- Extracted data from Wikipedia HTML tables
- Loaded response into lists transformed into data frame columns
- GitHub URL: https://github.com/Cameron-Nann-Python/ibm_data_science_capstone_rocket_launch_prediction/blob/main/step_1_data_collection/jupyter-labs-webscraping.ipynb



Data Wrangling

- Missing values and data types were assessed.
- The number of launches per launch site and orbit variations were determined.
- The number and type of mission outcomes were determined.
- A set of unsuccessful outcomes was made as conditional statement for a list comprehension of class labels (see notebook).
- The new column was added to the data frame that classified whether a launch outcome failed (0) or was successful (1).

- GitHub URL: https://github.com/Cameron-Nann-Python/ibm_data_science_capstone_rocket_launch_prediction/blob/main/step_2_data_wrangling/labs-jupyter-spacex-Data%20wrangling.ipynb

EDA with Data Visualization

- A scatterplot for payload mass vs flight number revealed the first stage is more successful with higher flight number volumes and larger payload masses
- A categorical scatterplot for launch site vs flight number showed the CCAFS SLC 40 launch site had the most flight numbers with successful first stages at higher flight numbers.
- A scatterplot between payload mass and orbit illustrated that low earth orbits with larger payloads resulted in more successful first stage launches.
- Numerical features were one-hot encoded for machine learning.
- GitHub URL: https://github.com/Cameron-Nann-Python/ibm_data_science_capstone_rocket_launch_prediction/blob/main/step_4_eda_python/edadataviz.ipynb

EDA with SQL

- The names of each launch site, booster versions that carried the maximum payload, and launch total launch outcomes were identified.
- Total payload mass for boosters launched by NASA (CRS) and average payload mass for booster version F9 v1.1 were identified.
- The date of the first successful landing outcome in ground pad was found.
- Boosters with successful drop ship landing outcomes and payload masses between 4000 and 6000 kg were noted.
- Landing outcomes from 2010-06-04 to 2017-03-20 were shown.
- GitHub URL: https://github.com/Cameron-Nann-Python/ibm_data_science_capstone_rocket_launch_prediction/blob/main/step_3_eda_sql/jupyter-labs-eda-sql-coursera_sqllite.ipynb

Build an Interactive Map with Folium

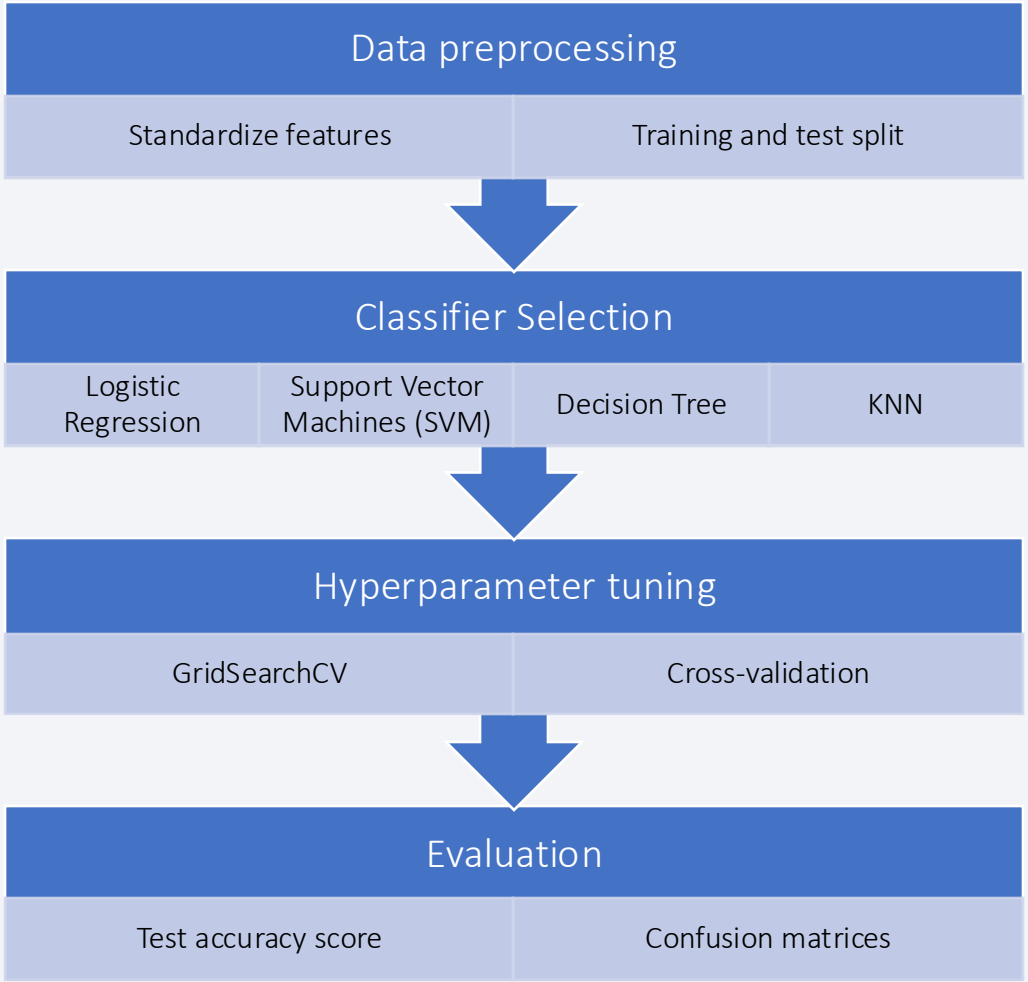
- The locations of launch sites were analyzed to determine optimal launch conditions.
- A folium map was implemented on an aggregated dataset of the launch sites. The average latitude and longitude coordinates were used to plot each launch site.
- It was observed all previous launch sites were situated near coastlines.
- A MarkerCluster group was added to the map to group each launch with it's outcome. Successful launches were designated with a green marker and unsuccessful launches were designated with a red marker.
- A distance function using MousePosition object to find distances between launch sites and proximal features like coastlines and highways. Transportation routes and safe crash zones were within 2 KM of the launch site.
- GitHub URL: https://github.com/Cameron-Nann-Python/ibm_data_science_capstone_rocket_launch_prediction/blob/main/step_5_visual_analytics_folium/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

- The pie chart and scatter chart slice to a specific launch site or show data for all launch sites based on dropdown input.
- The pie chart shows the launch sites ranked by the greatest number of successful launches based on launch site.
- The pie chart shows the launch outcome percentage for an individual launch site.
- The scatter chart shows launch outcomes by booster version category for a selected range input.
- GitHub URL: https://github.com/Cameron-Nann-Python/ibm_data_science_capstone_rocket_launch_prediction/blob/main/step_6_dashboard_dash_plotly/rocket_dashboard.py

Predictive Analysis (Classification)

- Features standardized with StandardScaler().
- Classifier objects used as estimators for GridSearchCV object for hyperparameter tuning.
- Cross-validation used over validation set with low sample size (<30).
- Decision tree classifier produced highest accuracy scores on training and test sets (0.93 and 0.83 respectively).
- GitHub URL: https://github.com/Cameron-Nann-Python/ibm_data_science_capstone_rocket_launch_prediction/blob/main/step_7_machine_learning/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb



Results

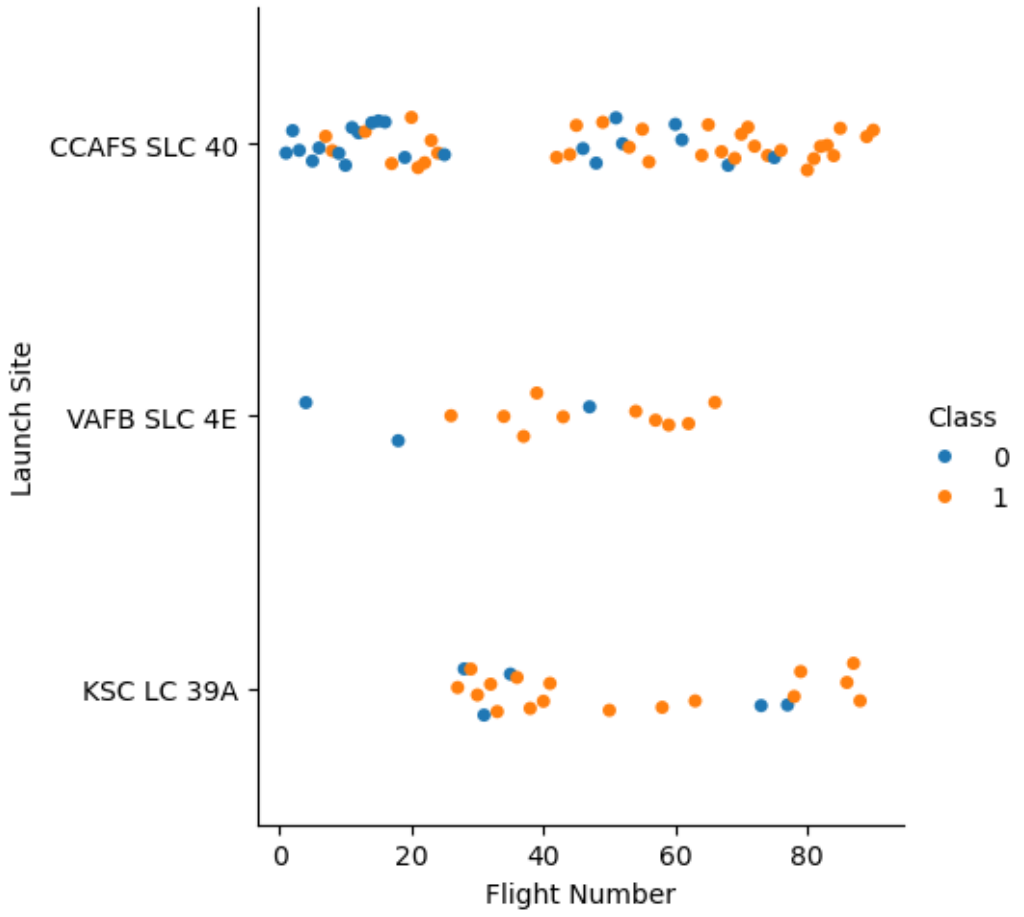
- Exploratory data analysis results
  - Successful launches appear to stop at certain payload thresholds. It can be seen that rockets launched by VAFB-SLC 4E have payloads that do not exceed 10,000 kg.
  - ES-L1, GEO, HEO, and SSO are tied for the highest success rate and SO has the lowest success rate. Launches seem to struggle with medium and low orbit paths.

- Predictive analysis results
  - The decision tree model performs the best with the highest accuracy on the training and test data sets.

Section 2

Insights drawn from EDA

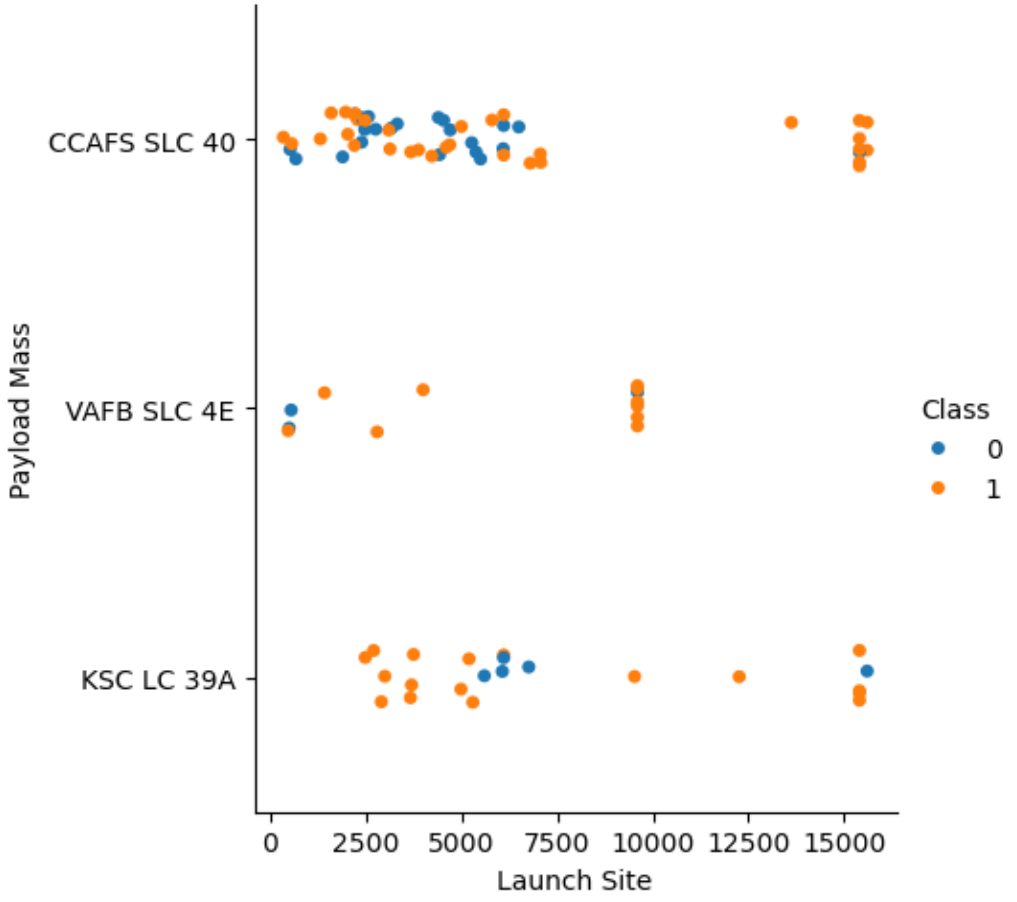
Flight Number vs. Launch Site



Now try to explain the patterns you found in the Flight Number vs. Launch Site scatter point plots.

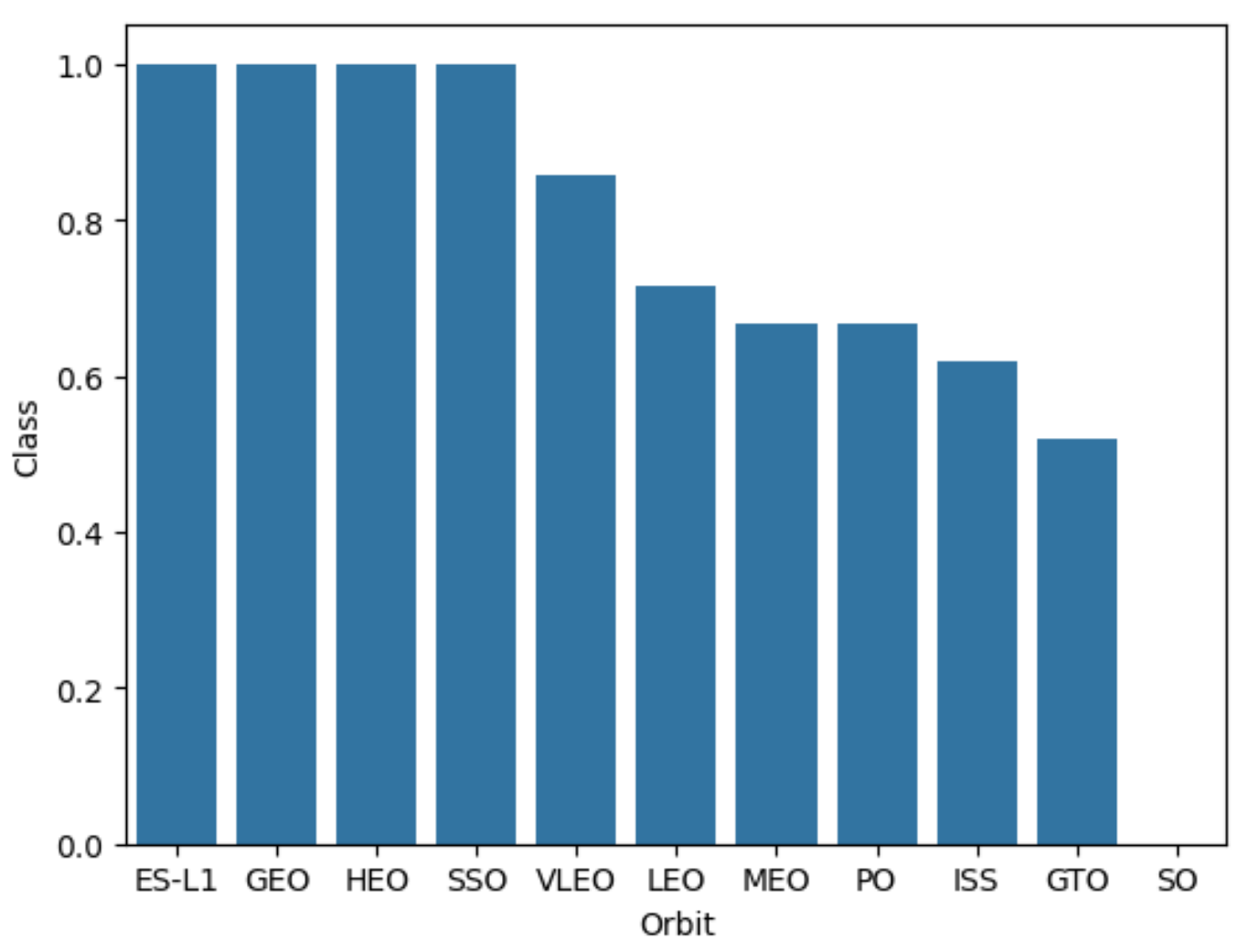
The CCAFS lauch site has the most failures with lower flight numbers. It appears the higher the flight number, the more likely it is for a successful launch.

Payload vs. Launch Site

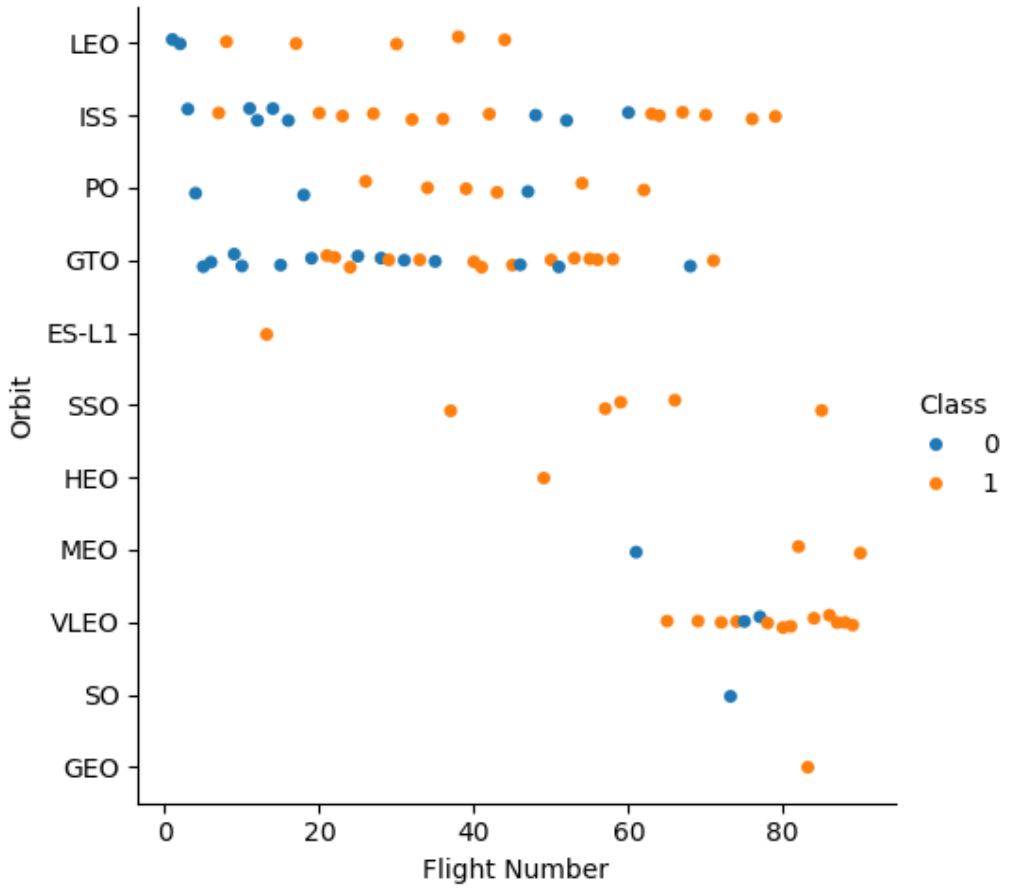


Now if you observe Payload Mass Vs. Launch Site scatter point chart you will find for the VAFB-SLC launchsite there are no rockets launched for heavypayload mass(greater than 10000).

Success Rate vs. Orbit Type

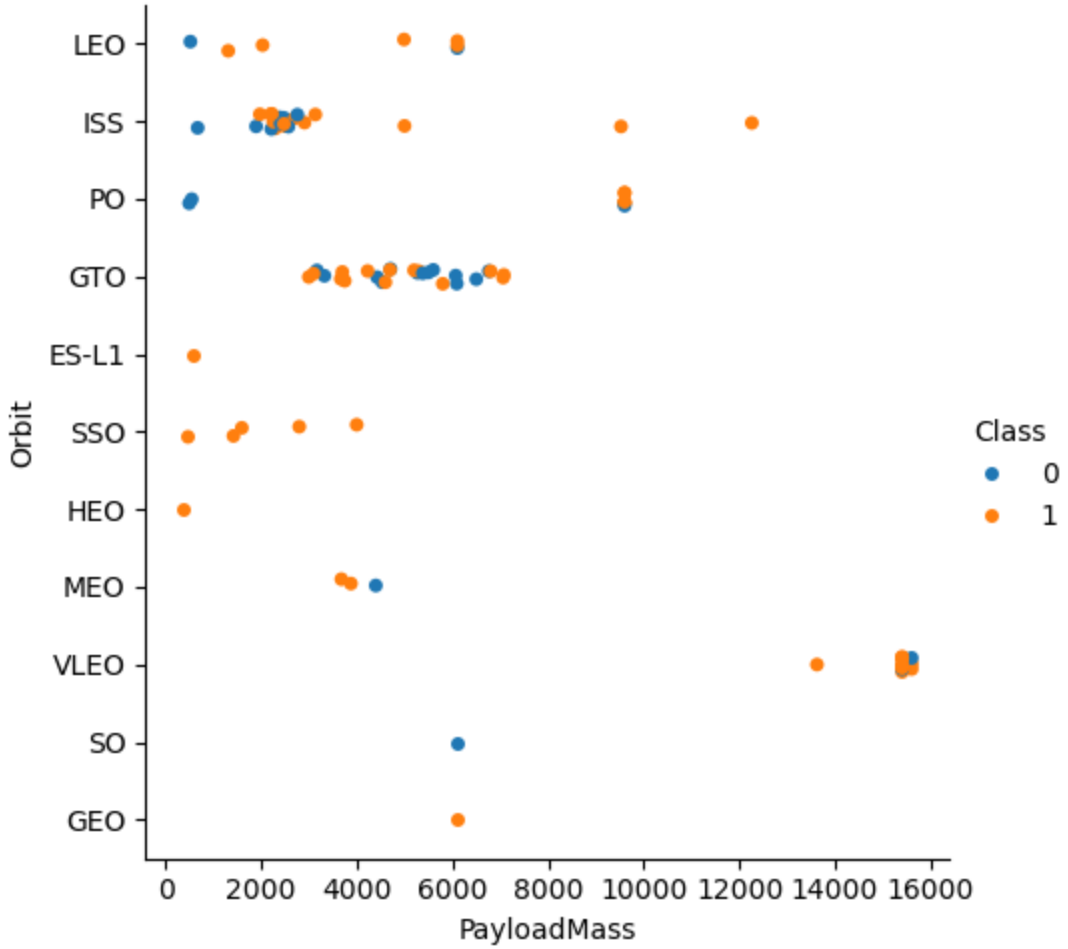


Flight Number vs. Orbit Type



You can observe that in the LEO orbit, success seems to be related to the number of flights. Conversely, in the GTO orbit, there appears to be no relationship between flight number and success.

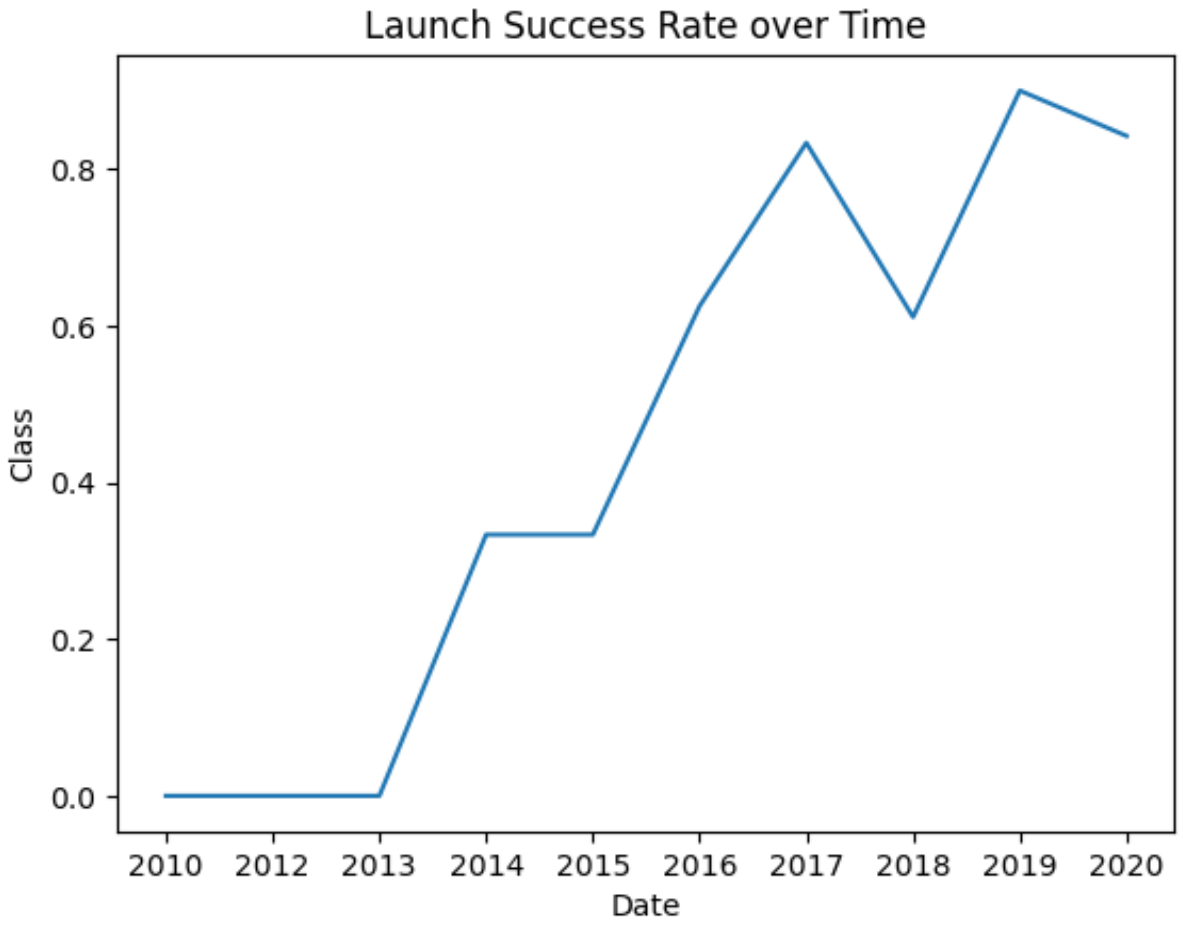
Payload vs. Orbit Type



With heavy payloads the successful landing or positive landing rate are more for Polar,LEO and ISS.

However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend



you can observe that the sucess rate since 2013 kept increasing till 2020

All Launch Site Names

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

Payload_Mass_KG_NASA_CRS
48213

Average Payload Mass by F9 v1.1

Avg_Payload_Mass_F9
6138.287128712871

First Successful Ground Landing Date

min(Date)
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

Mission_Successes	Mission_Failures
100	1

Boosters Carried Maximum Payload

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

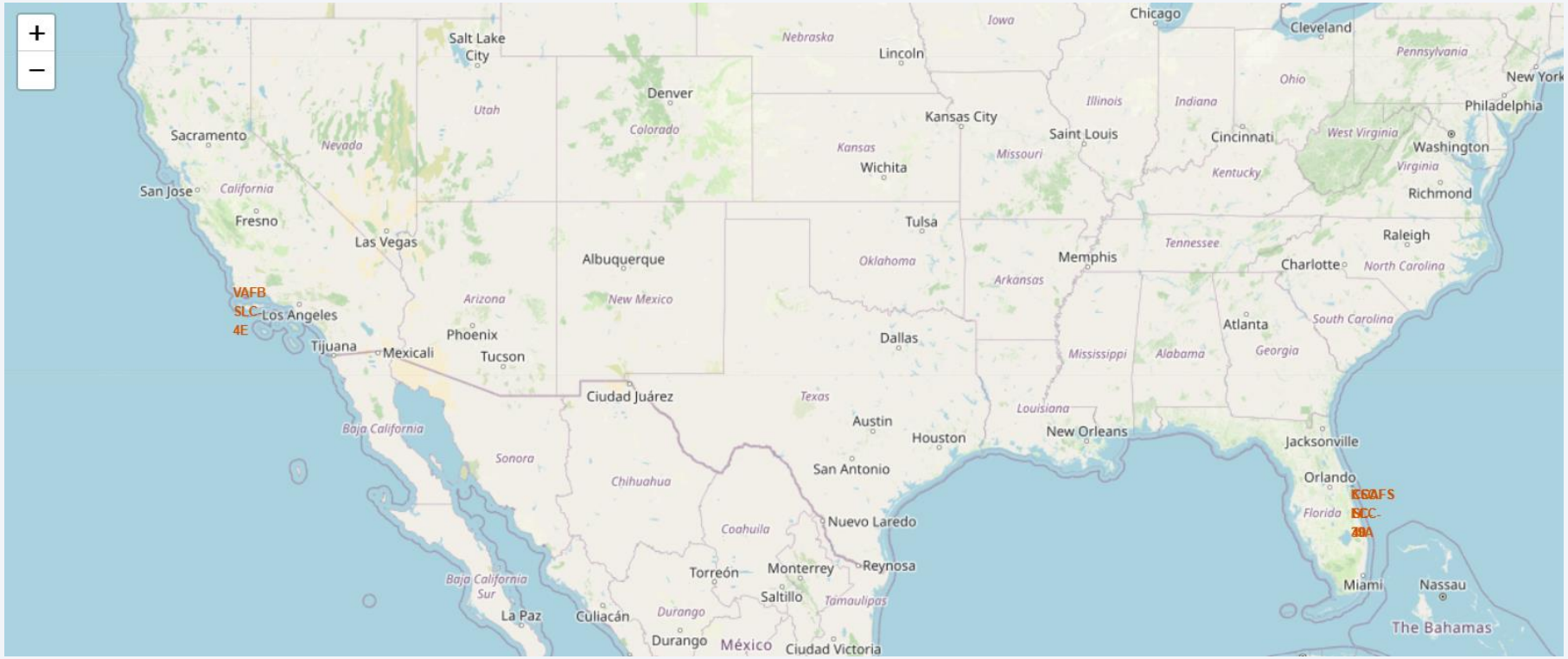
Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Date	Count	Landing_Outcome
2012-05-22	21	No attempt
2016-04-08	14	Success (drone ship)
2015-12-22	9	Success (ground pad)
2015-01-10	5	Failure (drone ship)
2014-04-18	5	Controlled (ocean)
2013-09-29	2	Uncontrolled (ocean)
2010-06-04	2	Failure (parachute)
2015-06-28	1	Precluded (drone ship)

Section 3

Launch Sites Proximities Analysis

Folium: Launch Sites Locations



- Launch sites are close to water so that the rocket can safely land in the water in case of failure.
- Rocket launches are not close to the equator as that is where the heat and humidity is maximized, which may jeopardize launch conditions.

Folium: Launch Site Outcomes

- The marker clusters highlight successful launches in green and failures in red.
- The KSC LC-39A launch site has a relatively high success rate compared to other sites.

Folium: Launch Site Proximities

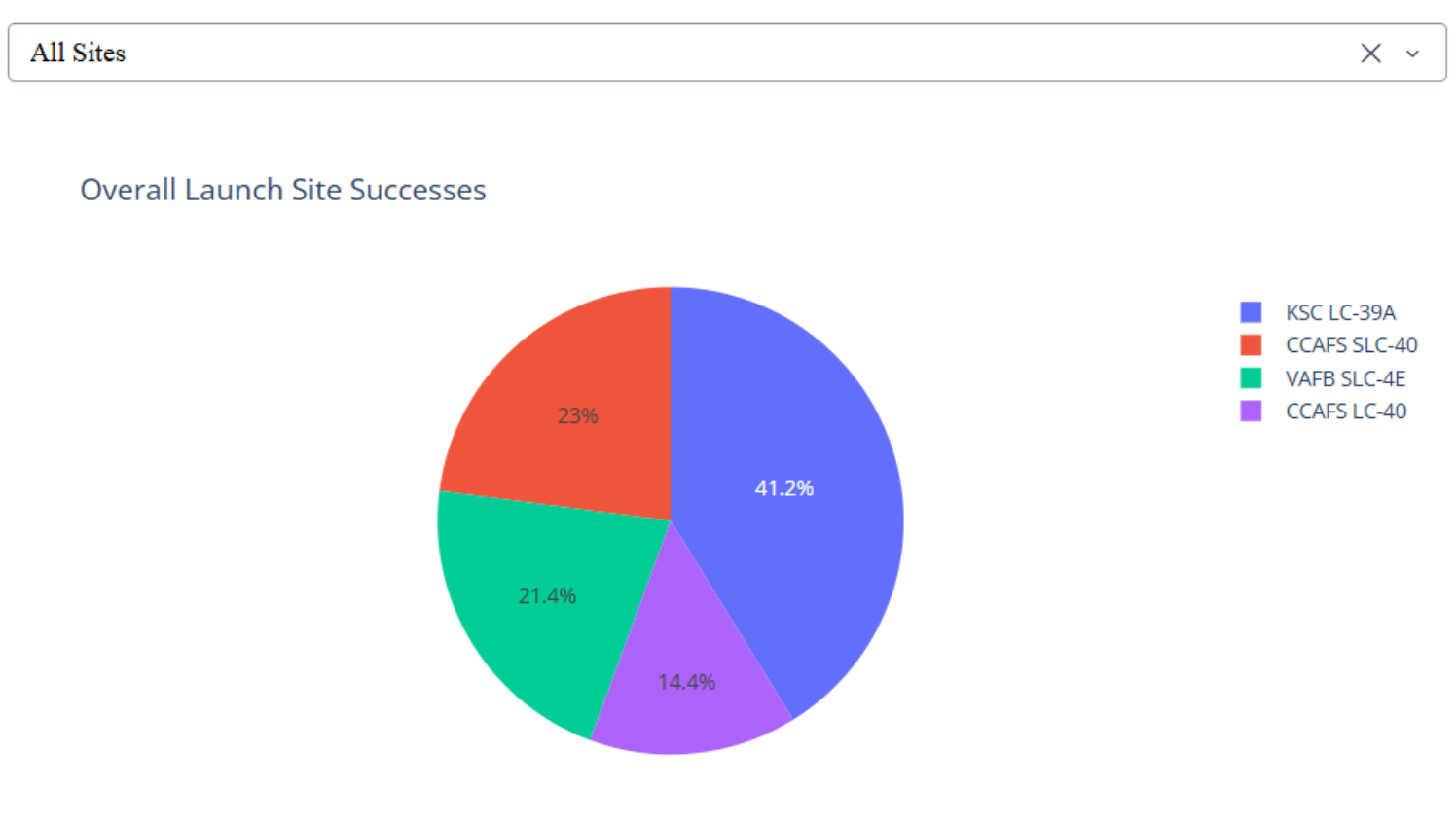
- The VAFB SLC-4E Launch site is 1.36 KM from the nearest coastline, which provides a safe way to land rockets away from cities.

Section 4

Build a Dashboard with Plotly Dash

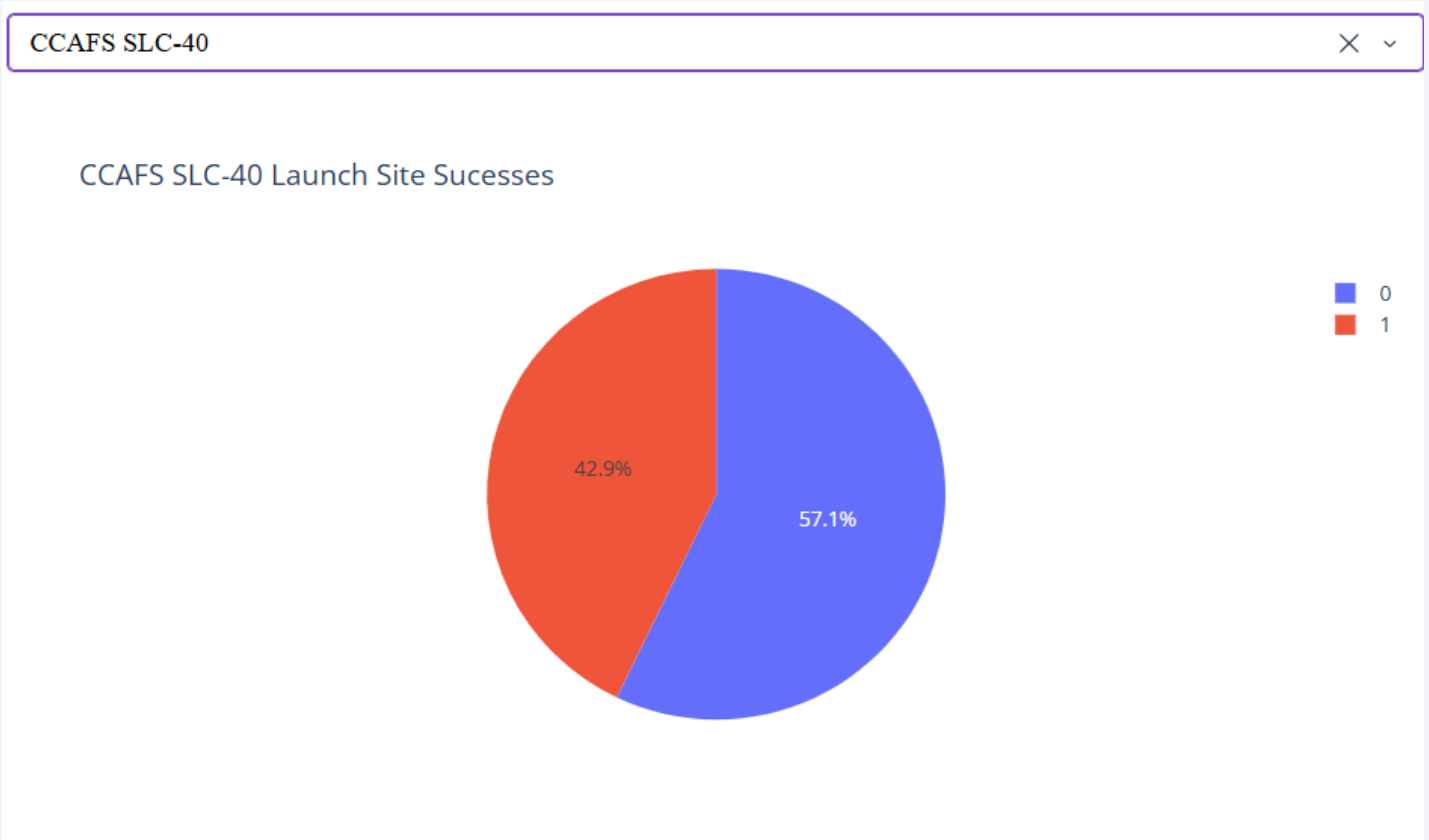
Plotly Dash Dashboard: Launch Success across Launch Sites

- KSC LC-39A has the largest number of launch site successes.
- CCAF LC-40 has the smallest number of launch site successes.



Plotly Dash Dashboard: Greatest Success Rate

- The CCAFS SLC-40 launch site has the greatest success rate at 42.9%.



Plotly Dash Dashboard: Payload vs Launch Outcome

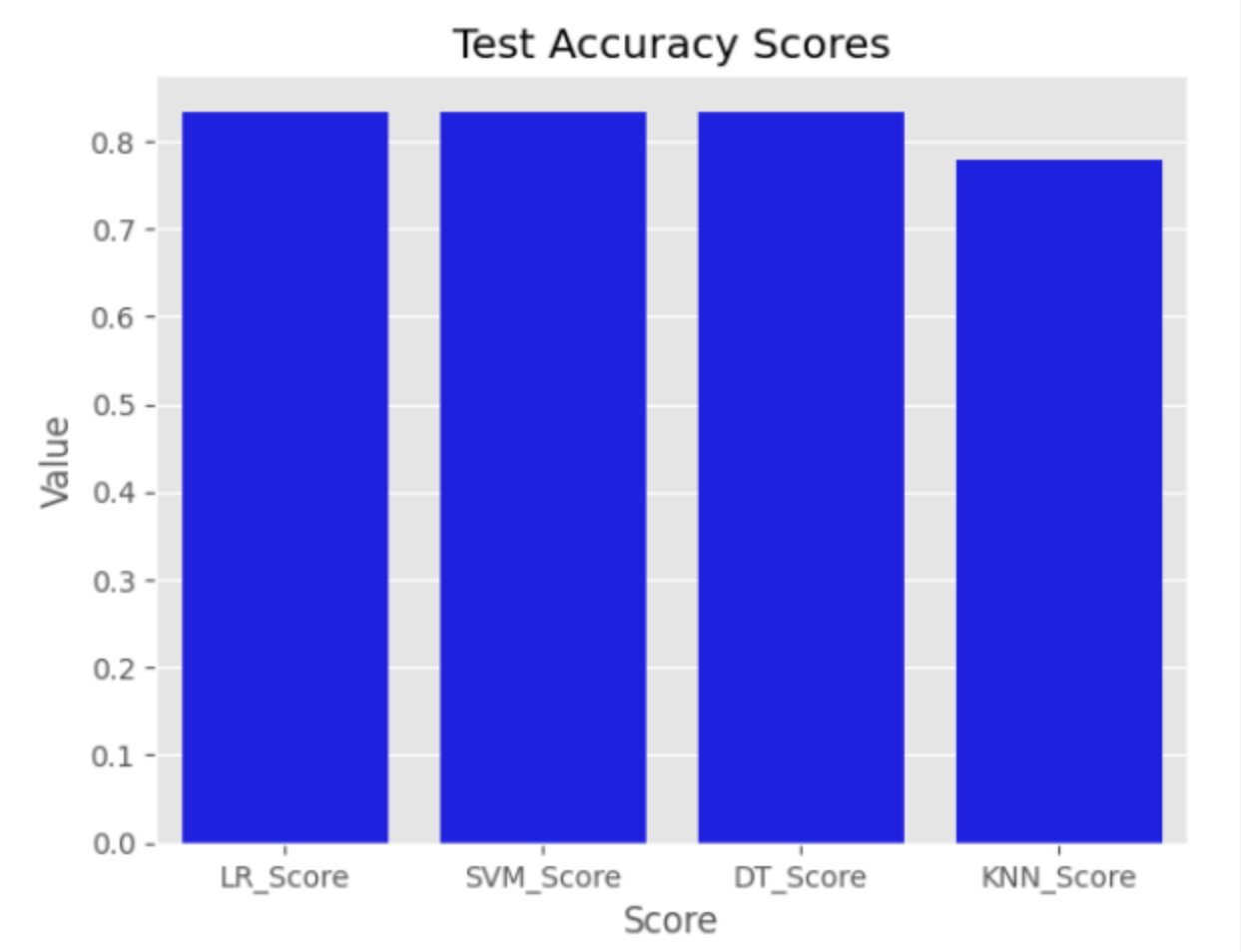
- The FT booster version category has the highest launch sucess rate

Section 5

Predictive Analysis (Classification)

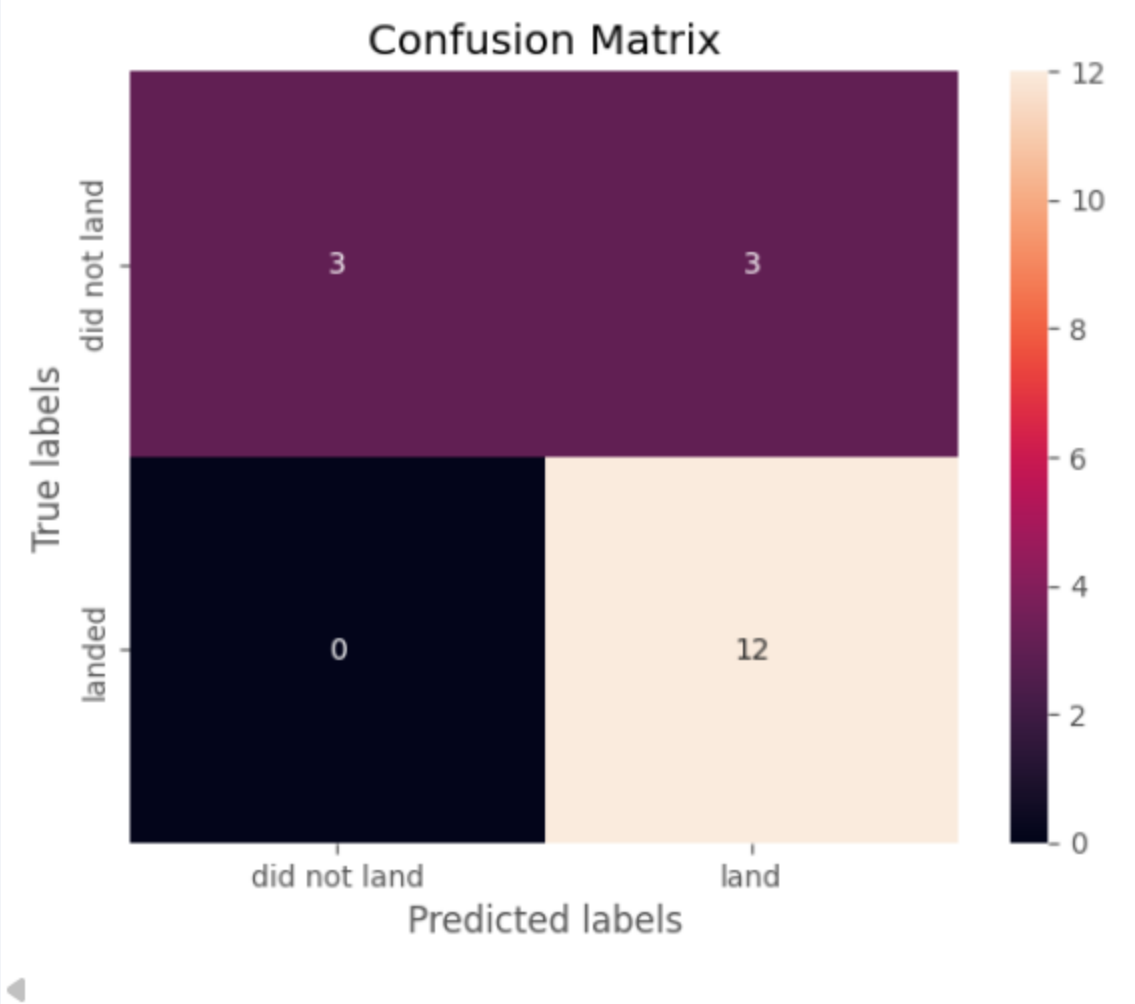
Classification Accuracy

- The decision tree classifier the highest training and test scores (0.9 and 0.83).



Confusion Matrix

- The decision tree classifier has 3 false positives and correctly found all other test classes.



Conclusions

- Successful launches are correlated with higher payload masses and greater flight number volumes.
- Launches appear to fail past certain orbit lengths and have payload limits based on the booster version.
- A decision tree classifier mostly accurately predicts a successful first stage launch.

Appendix: Launch Site Data Frame

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	Class
0	1	2010-06-04	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857	0
1	2	2012-05-22	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857	0
2	3	2013-03-01	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857	0
3	4	2013-09-29	Falcon 9	500.000000	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	B1003	-120.610829	34.632093	0
4	5	2013-12-03	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857	0

Appendix: Launch Record Data Frame

	FlightNumber	PayloadMass	Flights	Block	ReusedCount	Orbit_ES-L1	Orbit_GEO	Orbit_GTO	Orbit_HEO	Orbit_ISS	...	Serial_B1058	Serial_B1059	Serial_B1060	Serial_B1062	GridFins_False
0	1.0	6104.959412	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	1.0
1	2.0	525.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	1.0
2	3.0	677.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	1.0	...	0.0	0.0	0.0	0.0	1.0
3	4.0	500.000000	1.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	1.0
4	5.0	3170.000000	1.0	1.0	0.0	0.0	0.0	1.0	0.0	0.0	...	0.0	0.0	0.0	0.0	1.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
85	86.0	15400.000000	2.0	5.0	2.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	1.0	0.0	0.0
86	87.0	15400.000000	3.0	5.0	2.0	0.0	0.0	0.0	0.0	0.0	...	1.0	0.0	0.0	0.0	0.0
87	88.0	15400.000000	6.0	5.0	5.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0
88	89.0	15400.000000	3.0	5.0	2.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	1.0	0.0	0.0
89	90.0	3681.000000	1.0	5.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	1.0	0.0

Thank you!